RBS

The Responsibility Breakdown Structure (RBS) clarifies the responsibilities of each module and sub-task, ensuring that each team member's duties are clear, avoiding overlaps or ambiguities, thus optimizing project management. It helps the team define task objectives, enhance collaboration efficiency, and ensures that the project is completed on time and with high quality. Through RBS, the project can effectively monitor progress, make timely adjustments, and ensure that key modules receive sufficient attention, driving the smooth progress of the project. The project requirements are mainly focused on three aspects: large language model optimization, text processing features, and multi-role collaboration.

Large Language Model Optimization (LLM Optimization)

- **Model Training and Fine-tuning:** The project will first train the large language model and fine-tune its parameters to ensure the model better adapts to specific task requirements, improving generalization and task performance.
- **Inference Speed Optimization:** The project will focus on improving the model's inference speed by optimizing parallel processing mechanisms and resource management strategies, significantly increasing system response efficiency.
- **High Concurrency Processing Architecture Design:** A high-efficiency concurrency processing architecture will be designed, using distributed computing and load balancing technologies to ensure the system runs stably in high-concurrency scenarios and can efficiently handle large volumes of requests.

Text Processing Features

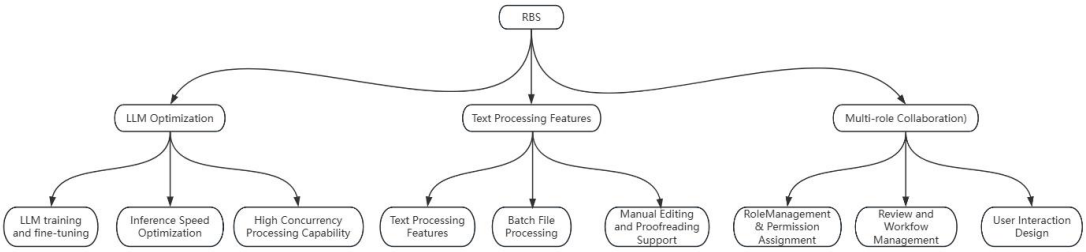
- **Title and Summary Generation:** The project will develop extractive title and summary generation functions, designing efficient title generation algorithms and summary extraction algorithms to automatically extract key information from text and generate concise and accurate titles and summaries.
- **Batch File Processing:** A batch file processing module will be designed to support batch processing of multiple file formats, significantly improving the efficiency of large-scale text processing.
- **Manual Editing and Proofreading Support:** Manual editing and proofreading support features will be provided, along with a user-friendly interface that allows users to perform manual editing and proofreading to ensure the accuracy and completeness of text content.

Multi-role Collaboration

- **Role Management and Permission Assignment:** A flexible role management

system will be designed, supporting the configuration of different permission levels to ensure that roles can access and operate system functions based on their responsibilities.

- **Review Process Functionality:** The project will develop a review process feature that supports multi-round approval mechanisms, ensuring efficient management of workflows and accurate execution of tasks.
- **User Interaction Design:** Emphasis will be placed on user interaction design, creating intuitive and user-friendly interfaces to optimize the experience for multi-role users and enhance team collaboration efficiency and smoothness.



WBS

The Work Breakdown Structure (WBS) is a method of breaking down a project into smaller, manageable work units. It helps the project team refine specific work tasks and clarify each stage and step of the project. The WBS in this project covers the following core work areas, which are derived from the lowest-level requirements in the Requirements Breakdown Structure (RBS):

Large Language Model Optimization (LLM Optimization)

Model Training and Fine-tuning:
The project will first train the large language model and fine-tune its parameters to ensure the model better adapts to specific task requirements, improving generalization and task performance.

- Train the large model
- Fine-tune parameters
- Improve task adaptability
- Review: Evaluate model performance and adjust hyperparameters
- Test: Test model generalization on different tasks

Inference Speed Optimization:
The project will focus on improving the model's inference speed by optimizing parallel processing mechanisms and resource management strategies, significantly increasing system response efficiency.

- Optimize parallel processing
- Optimize resource management

- Improve system response efficiency
- Review: Monitor optimization effects and analyze bottlenecks
- Test: Conduct stress testing to validate optimization results

High Concurrency Processing Architecture Design:
A high-efficiency concurrency processing architecture will be designed, using distributed computing and load balancing technologies to ensure the system runs stably in high-concurrency scenarios and can efficiently handle large volumes of requests.

- Design concurrency architecture
- Utilize distributed computing
- Optimize load balancing
- Review: Evaluate architecture design and adjust strategies
- Test: Simulate high-concurrency scenarios to test system stability

Text Processing Features

Title and Summary Generation:
The project will design extractive title and summary generation functions, creating efficient title generation and summary extraction algorithms to automatically extract key information and generate concise, accurate titles and summaries.

- Design efficient title generation algorithms
- Develop summary extraction algorithms
- Automatically extract key information from text
- Generate concise and accurate titles and summaries
- Review: Evaluate the quality of generated titles and summaries, optimize algorithms
- Test: Test accuracy and readability across different text samples

Batch File Processing:
The project will implement a batch file processing module to support multiple file formats, significantly enhancing the efficiency of large-scale text processing.

- Design a batch file processing module
- Support batch processing of multiple file formats
- Improve efficiency in large-scale text processing
- Review: Assess the stability and compatibility of batch processing
- Test: Conduct large-scale file processing tests to evaluate system performance

Manual Editing and Proofreading Support:
The project will provide manual editing and proofreading support with a user-friendly interface, allowing users to manually edit and proofread text to ensure accuracy and completeness.

- Provide manual editing and proofreading support features
- Design a user-friendly editing interface
- Enable users to manually edit and proofread text
- Ensure text accuracy and completeness
- Review: Evaluate the usability of the user interface and editing functions
- Test: Conduct user testing to ensure the effectiveness of editing and proofreading features

Multi-role Collaboration

Role Management and Permission Assignment:

A flexible role management system will be designed to support the configuration of different permission levels, ensuring that roles can access and operate system functions based on their responsibilities.

- Design a flexible role management system
- Configure different permission levels
- Ensure roles can access system functions based on responsibilities
- Review: Evaluate role configurations and permission settings
- Test: Verify access control and permission enforcement across different roles

Review Process Functionality:

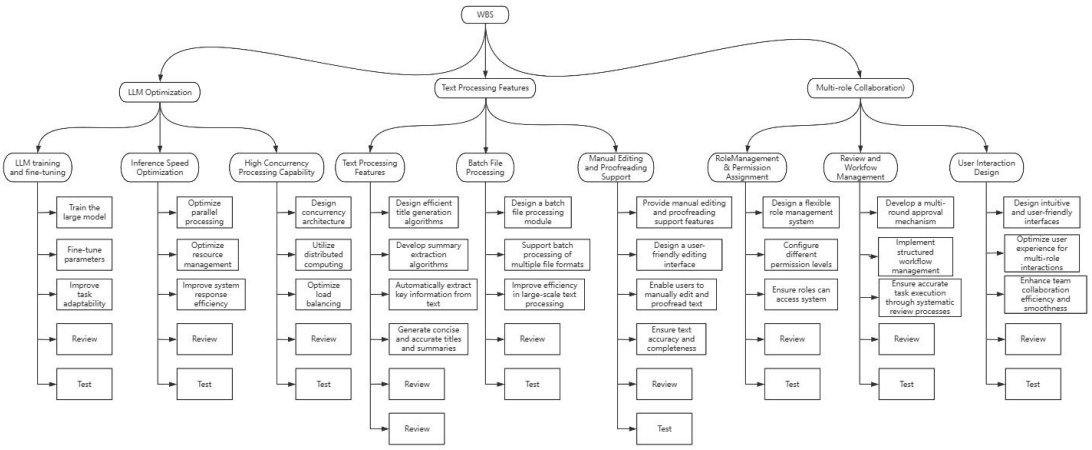
The project will develop a review process feature supporting multi-round approval mechanisms to ensure efficient workflow management and accurate task execution.

- Develop a multi-round approval mechanism
- Implement structured workflow management
- Ensure accurate task execution through systematic review processes
- Review: Assess workflow efficiency and approval accuracy
- Test: Simulate multi-stage approval processes and validate functionality

User Interaction Design:

The project will focus on user interaction design, creating intuitive and user-friendly interfaces to enhance the experience for multi-role users and improve collaboration efficiency.

- Design intuitive and user-friendly interfaces
- Optimize user experience for multi-role interactions
- Enhance team collaboration efficiency and smoothness
- Review: Analyze usability and user satisfaction
- Test: Conduct user testing to refine interaction flow and interface usability



Gantt Chart

The development process for the Zhiwen Intelligent Writing Platform is organized into several distinct phases, each focusing on a key aspect of the project lifecycle:

Planning (2025-03-24 to 2025-03-31):

The project kicks off with a thorough requirements gathering phase. During this period, the team collaborates to define project objectives, scope, and essential features, establishing a solid foundation for all subsequent work.

System Design (2025-03-31 to 2025-04-08):

In this phase, the overall system architecture is designed. Decisions regarding the technology stack, infrastructure requirements, and system components are made to ensure that the platform can support high-concurrency operations and large-scale data processing.

Module Development (Starting 2025-04-08):

Multiple modules are developed concurrently to expedite the project:

- Intelligent Text Generation Module (2025-04-08 to 2025-05-05): Utilizes GPT-2 to generate high-quality, long-form text with multiple output options.

- Summary Extraction Module (2025-04-08 to 2025-05-05): Implements techniques to extract key content and produce concise summaries.
- Batch Text Processing Module (2025-04-15 to 2025-05-05): Handles large volumes of file uploads (Word, TXT, ZIP) and automates text processing tasks.
- Manual Editing & Review Module (2025-04-22 to 2025-05-10): Provides tools for post-generation editing and quality assurance.
- Multi-Role Management & Review Workflow (2025-04-22 to 2025-05-10): Establishes collaborative processes to support various user roles (e.g., content creators, reviewers).

Infrastructure Development (2025-04-08 to 2025-05-15):

The backend and frontend of the platform are built in parallel:

- Backend Development: Focuses on implementing GPU acceleration and distributed deployment strategies using technologies like Flask, MySQL, and Redis.
- Frontend Development: Uses Vue.js and Ant Design to create an intuitive and user-friendly interface.

Integration and Testing (2025-05-06 to 2025-05-25):

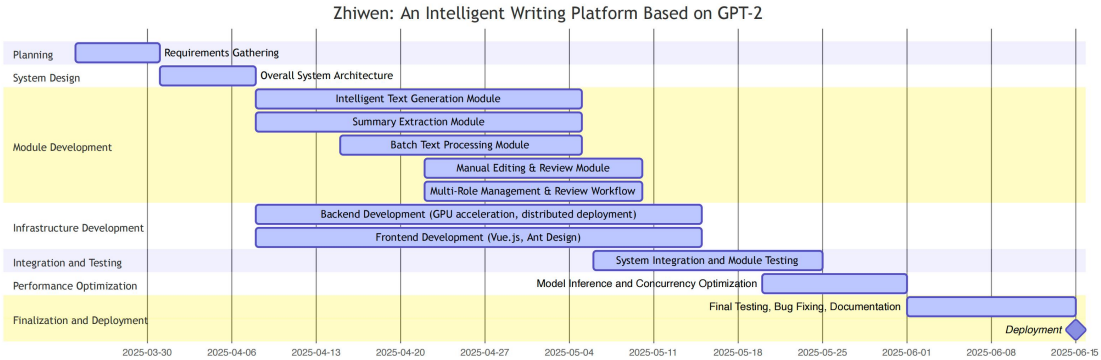
All modules are integrated and tested to ensure seamless interaction and system stability. This phase is crucial for identifying and resolving any inconsistencies or bugs before moving forward.

Performance Optimization (2025-05-20 to 2025-06-01):

Special attention is given to enhancing the platform’s performance by optimizing model inference and ensuring the system can handle high levels of concurrency effectively.

Finalization and Deployment (2025-06-01 to 2025-06-15):

The project concludes with final testing, bug fixing, and thorough documentation. Once these tasks are completed and verified, the platform is deployed, marking the successful launch on June 15.

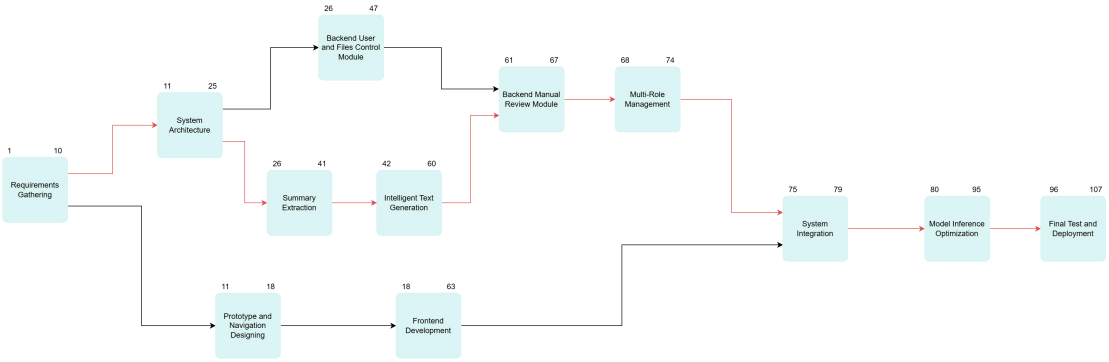


Critical Path Diagram

This critical path diagram meticulously illustrates the essential sequence of activities in the development and deployment of our web platform, revealing the dependencies between various project phases. Initially, the project commences with the requirements gathering phase, where the functional needs and objectives of the platform are clearly defined. This is followed by the system architecture design phase, which lays the foundation for the platform's overall structure. Subsequently, the process moves to the summary extraction phase, providing the core data support necessary for the platform.

The diagram further details the intelligent title generation phase, a crucial component for content generation on the platform, aiding in the dynamic creation of high-quality titles. Following this, the project progresses to the development of the backend user and file control modules, ensuring efficient management of user data and files. This is closely followed by the design and implementation of the backend manual review module, facilitating a smooth content review process. The platform's multi-role management module is also developed during this phase, ensuring personalized user experiences for different roles.

After completing the development of these key modules, the project enters the system integration phase, where the front-end, middle-end, and back-end systems are integrated to ensure seamless collaboration. Next, through model inference optimization, the platform's performance and efficiency are enhanced, ensuring stable operation even under high load conditions. Finally, the final testing and deployment phase involves comprehensive testing of the platform to ensure all functionalities work as expected, preparing the platform for launch.

This critical path diagram not only clearly demonstrates the temporal dependencies between each phase but also helps us identify the shortest path to

project completion, providing robust support for rational project scheduling and timely delivery. Through this systematic path management, we can efficiently advance the development and deployment of the platform.